

Iris Yin

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EDUCATION

Carnegie Mellon University, B.S. in Statistics and Machine Learning

Expected May 2028

Relevant Coursework: Data Structures & Algorithms (C), Fundamentals of Programming (Python), Methods of Statistics and Data Science (R), Introduction to Deep Learning

TECHNICAL SKILLS

Programming Languages: SQL, Python, C/C++, JavaScript/HTML, Java, R

Frameworks & Tools: LLM, AWS (EC2, IAM, Secrets Manager), Kubernetes, PyTorch, Terraform, Docker, GitHub Actions / CI-CD, Git, Modal, Slack & Notion APIs, Agent architecture, Claude Code, Pandas, Scikit-learn, HuggingFace, Weights & Biases (W&B), MCP tool integration, hybrid RAG (vector search + BM25 + reciprocal rank fusion), model fine-tuning, evaluation & benchmarking pipelines, Swift

WORK EXPERIENCE

Poseidon.ai

Palo Alto, CA

AI Engineering Intern | Python, AWS, Terraform, Docker, GPT-4o mini, MCP

May 2026 - Aug 2026

- **AI Agent development**
 - Engineered a Slack codebase Q&A agent using the Hermes Agent framework deployed on AWS EC2 with zero open inbound ports, utilizing a hybrid vector + BM25 reciprocal rank fusion retrieval pipeline
 - Deployed and debugged services on an existing Kubernetes cluster
 - Led feature releases through the full software engineering lifecycle: Feature Complete (FC), Code Freeze, Beta testing, internal dogfooding, and Release Candidate (RC) launch.
 - Owned and redesigned an engineering activity platform (Java backend for PR processing, JavaScript frontend) for a 26-engineer organization, transforming unstructured raw Markdown feeds into structured HTML digests praised by leadership.
- **AI Speech Model Optimization**
 - **Fine-tuned Whisper-medium with LoRA** on company-owned Hindi speech corpus, improving word-error-rate (WER) on 95.8% of 2,334 held-out recordings (~50-point median gain) and **cutting out-of-domain WER from 68.5% to 17.8%**.
- **CI/CD Workflow**
 - Built containerized CI/CD pipelines with GitHub Actions and Docker, using Claude Code to scaffold workflows, then optimized image builds and runner infrastructure to cut deployment time 40% and eliminate manual configuration errors.
- Designed the architecture diagrams for the launch blog, iterating A/B versions into the one adopted for release — co-authored the project's VANI blog as a result.

Adulding101 - CMU Human Computer Interaction Institute Spud Lab

Pittsburgh, PA

Research Assistant | Swift, JavaScript

Jan 2026 - Present

- Contributed to a behavioral-research iOS app studying habit formation across cybersecurity and general life-skills interventions (solo vs. paired-accountability conditions); integrated a 1,000+ image asset catalog (688 image sets across skin-tone, overlay, and mirrored-state variants)
- Refined SwiftUI layout/alignment for the avatar and celebration-illustration system shared across all study conditions.

PROJECTS

Emotion Classifier | PyTorch, HuggingFace Transformers, Python

Sep 2026

- Fine-tuned DistilBERT on a 16K-tweet, 6-class emotion dataset, reaching 0.92 macro-F1 and 94% accuracy on validation, up from a 0.09 macro-F1 majority-class baseline; chose macro-F1 as the headline metric so rare emotions counted equally.
- Stress-tested the model with adversarial inputs (sarcasm, negation, emotions about other people) and traced per-class errors, such as love being confused with joy, to class imbalance and noisy hashtag-derived labels

ScottyLabs NOVA Hackathon - Nominated for 1st Place | Python, XGBoost, Scikit-learn

Nov 2025

FetoScope

- Built an XGBoost model called FetoScope to predict fetal health from clinical features with a four person team.
- Led hyperparameter tuning for model, improving prediction accuracy from 83% to 94.6%.
- Evaluated model robustness under real-world clinical deployment constraints and presented technical findings to engineering judges.

Text Mining & Sentiment Analysis - Reddit Relationships Dataset | R, tidytext, NLP

Oct 2025

- Analyzed 10,000 Reddit posts using NLP techniques in R (tidytext), extracting insights on gender differences in online communication patterns.
- Implemented a text preprocessing pipeline (tokenization, stemming, stopword removal, TF-IDF weighting) to identify key thematic signals.
- Formulated and tested 3 hypotheses using proportion-based statistical hypothesis testing, identifying 95% overlap in relationship concerns across genders.